

CSE 815: Machine Learning

Week 1, Part 2 — Research Articles & Mathematical Foundations

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Today's Agenda

- 1 Research Paper 1 — Samuel (1959): The Birth of ML
- 2 Research Paper 2 — Jordan & Mitchell (2015)
- 3 Mathematical Formulation of the Learning Problem
- 4 Case Study — Spam Filtering: The Classic ML Success
- 5 Guided Discussion & Live Exercise

Research Articles Covered

- 1 Samuel, A. L. (1959). *Some Studies in Machine Learning Using the Game of Checkers*.
- 2 Jordan, M. I., & Mitchell, T. M. (2015). *Machine Learning: Trends, Perspectives, and Prospects* (Science).

Case Study

The Spam Filtering Problem — The canonical industrial success story of supervised learning.

CLO Addressed

CLO1 — Understand fundamental concepts
CLO3 — Explain mathematical principles
PLO(a), PLO(b)

Research Paper 1 — Samuel (1959): The Birth of ML

Samuel (1959): The Checkers Program

Full Citation

Samuel, A. L. (1959). **Some Studies in Machine Learning Using the Game of Checkers.** *IBM Journal of Research and Development*, 3(3), 210–229.

Key Contributions

- First computer program that **learned** from experience rather than following fixed rules
- Used **minimax search** with an **evaluation function** (position scoring)
- Self-play: played thousands of games against itself to improve
- Achieved **competitive play** against strong human players by 1962
- **Popularised** the term “machine learning” through the paper’s title

The Key Insight

Samuel’s program **learned** its evaluation weights:

- 1 Initial weights hand-coded by Samuel
- 2 Program played against itself
- 3 Weights adjusted to **maximize** winning outcomes
- 4 Over time, the program discovered strategies that Samuel had not explicitly programmed

This was the first demonstration of a machine improving with experience.

Samuel (1959): Why This Paper is Seminal

The Core Paradigm Samuel Established

- 1 Define a **performance measure** (win rate against an opponent)
- 2 Provide **experience** (self-play games)
- 3 Let the system **adjust its own parameters** to improve on that measure

This pre-dates Mitchell's formal definition by nearly 40 years — Samuel *demonstrated* the idea that Mitchell later *formalised*.

Samuel's Checkers Player in Mitchell's T-P-E Framework (recall Part 1)

Component	Instantiation in Samuel (1959)
Task T	Playing checkers: select a move from the current board position
Performance P	Fraction of games won against an opponent
Experience E	Games played against itself, plus recorded master games

Research Paper 2 — Jordan & Mitchell (2015)

Jordan & Mitchell (2015): Machine Learning as a New Engineering Discipline

Full Citation

Jordan, M. I., & Mitchell, T. M. (2015). **Machine Learning: Trends, Perspectives, and Prospects.** *Science*, 349(6245), 255–260.

Core Thesis

Machine learning is emerging as a **new engineering discipline** that:

- Bridges **computer science** (algorithms, systems)
- Bridges **statistics** (inference, uncertainty)
- Bridges **optimization** (gradient methods, convexity)
- Is **transforming science** and industry

The “New Engineering” Framing

ML is increasingly deployed where the **cost of an error is high** — autonomous vehicles, medical diagnosis, financial trading. Jordan & Mitchell argue this places new demands on the field:

- **Robustness** to distribution shift
- **Explainability** of decisions
- **Fairness** and non-discrimination
- **Safety** in open-world settings

(paraphrased from Jordan & Mitchell, 2015)

Jordan & Mitchell (2015): Key Trends Identified

Trend	Description	Implication
Scale	Big data + cloud computing	Algorithms must scale to billions of examples
Deep Learning	Hierarchical feature learning	End-to-end learning reduces feature engineering
Transfer Learning	Knowledge across tasks	Pre-trained models reduce data requirements
Interpretability	Understanding model decisions	Critical for high-stakes domains (health-care, finance)
Privacy	Learning without exposing data	Differential privacy; federated learning (post-2016)
Causality	From correlation to causation	Essential for decision-making systems

The Paper's Thesis

The integration of ML with **domain science** (biology, physics, economics) is the frontier. ML is no longer just a tool for tech companies—it is a **foundational methodology** for all of science.

Mathematical Formulation of the Learning Problem

The Formal Setup: A Mathematical View

The Components

- **Input space:** $\mathcal{X}$ — the set of all possible inputs
- **Output space:** $\mathcal{Y}$ — the set of all possible outputs
- **Data distribution:** $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$ (unknown, fixed)
- **Hypothesis space:** $\mathcal{H}$ — set of functions $h : \mathcal{X} \rightarrow \mathcal{Y}$ we can learn
- **Loss function:** $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}^+$ — measures prediction error

Example: Binary Classification

- $\mathcal{X} = \mathbb{R}^d$ (feature vectors)
- $\mathcal{Y} = \{0, 1\}$ (spam / not spam)
- $\ell(y, \hat{y}) = 1[y \neq \hat{y}]$ (0-1 loss)
- $\mathcal{H} = \{h(x) = \sigma(w^T x + b)\}$ (logistic regression family)

The Learning Objective

We want to find the hypothesis $h \in \mathcal{H}$ that **minimises the expected loss** over the true data distribution:

$$R_{\text{true}}(h) = \mathbb{E}_{(x,y) \sim \mathcal{D}} [\ell(h(x), y)]$$

This is called the **True Risk** (or generalisation error). **Problem:** $\mathcal{D}$ is unknown. We only have a finite training set:

$$\mathcal{S} = \{(x_1, y_1), \dots, (x_n, y_n)\} \sim \mathcal{D}^n$$

Empirical Risk Minimization (ERM)

The Empirical Risk

Since we cannot compute R_{true} , we approximate it with the **Empirical Risk** (average loss on training data):

$$R_{\text{emp}}(h) = \frac{1}{n} \sum_{i=1}^n \ell(h(x_i), y_i)$$

The learning algorithm selects:

$$h^* = \arg \min_{h \in \mathcal{H}} R_{\text{emp}}(h)$$

This is **Empirical Risk Minimization (ERM)**.

Example: Linear Regression

$$h(x) = w^T x, \ell(y, \hat{y}) = (y - \hat{y})^2$$

ERM becomes:

$$w^* = \arg \min_w \frac{1}{n} \sum_{i=1}^n (y_i - w^T x_i)^2$$

Which is exactly the **least squares** problem.

The Generalization Gap

We care about $R_{\text{true}}(h^*)$, not $R_{\text{emp}}(h^*)$.

- If R_{emp} is low but R_{true} is high $\rightarrow$ **overfitting**
- If R_{emp} is high $\rightarrow$ **underfitting** (or poor hypothesis class)

Learning theory (Vapnik-Chervonenkis, 1971) bounds the generalization gap:

$$R_{\text{true}}(h) \leq R_{\text{emp}}(h) + \text{complexity}(\mathcal{H}, n)$$

More complex $\mathcal{H}$ (e.g., neural networks) require more data to keep the gap small.

The Bias-Variance-Noise Decomposition (Mathematical)

The Fundamental Decomposition

For regression with squared loss $\ell(y, \hat{y}) = (y - \hat{y})^2$, the **expected prediction error** at a point x decomposes as:

$$\mathbb{E}[(y - \hat{y})^2] = \text{Bias}^2 + \text{Variance} + \text{Noise}$$

- **Bias:** Error from approximating the true function with a limited hypothesis class:

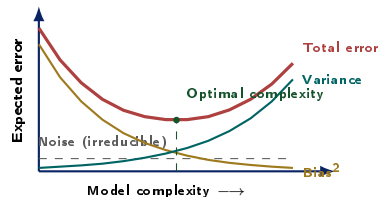
$$\text{Bias}^2 = (\mathbb{E}[\hat{h}(x)] - \mathbb{E}[y|x])^2$$

- **Variance:** Error from sensitivity to the particular training sample:

$$\text{Variance} = \mathbb{E}[(\hat{h}(x) - \mathbb{E}[\hat{h}(x)])^2]$$

- **Noise:** Irreducible error from the data generation process:

$$\text{Noise} = \text{Var}(y|x)$$



Key Insight: As complexity grows, Bias^2 falls but Variance rises. Their sum is **U-shaped**; the best model sits at its minimum. No model can beat the noise floor.

The Hypothesis Space and Capacity

Hypothesis Space $\mathcal{H}$

The set of all functions the learning algorithm can choose from. It encodes our **inductive bias** — the assumptions that let a learner generalise beyond the training set at all.

Algorithm	Hypothesis Space	Inductive Bias
Linear Regression	$\{h(x) = w^T x \mid w \in \mathbb{R}^d\}$	Linear relationship
Decision Tree	$\{h \text{ is a tree of depth } \leq k\}$	Hierarchical, axis-aligned splits
SVM (RBF Kernel)	$\{h(x) = \sum_i \alpha_i K(x_i, x) + b\}$	Smooth functions in RKHS
Neural Network	$\{h(x) = \text{composition of layers}\}$	Compositional hierarchy

Why This Matters

Choosing an algorithm *is* choosing an inductive bias. A linear model cannot represent an XOR boundary no matter how much data you give it — the failure is in $\mathcal{H}$, not in the optimiser.

Case Study — Spam Filtering: The Classic ML Success

The Spam Filtering Problem

The Task

Classify incoming email as **spam** or **ham** (legitimate).

- **Input:** Raw email text + metadata (sender, headers)
- **Output:** $y \in \{\text{spam}, \text{ham}\}$
- **Experience:** Millions of labeled emails
- **Performance:** Accuracy, precision, recall, F1-score

Why It Was a Breakthrough

- Pre-ML approach: hand-coded rules (e.g., “contains ‘lottery’ → spam”)
- Problem: spammers constantly adapt their tactics
- Rule-based filters required **thousands** of rules and constant maintenance
- ML approach: automatically learns new patterns from labeled data

Features Used in Spam Filters

- Word frequencies (bag-of-words)
- N-gram presence (e.g., “free”, “\$\$\$”)
- Sender domain reputation
- Email header metadata
- HTML content structure
- Character-level patterns (e.g., “l1ttery” evasion)
- Metadata: time of day, recipient history

Typical feature vector dimension:

10,000–100,000 sparse binary features (presence of specific words/patterns).

Spam Filtering: The ML Approach

Algorithmic Evolution

Era	Algorithm
Early 2000s	Naïve Bayes (probabilistic)
Mid 2000s	SVM (linear kernel)
Late 2000s	Logistic Regression + feature engineering
Early 2010s	Random Forests, Gradient Boosting
Present	Ensembles + deep models for context

Why Naïve Bayes Worked Surprisingly Well

Despite the **independence assumption** (words in an email are not truly independent), Naïve Bayes performs well because:

- 1 It is highly **robust** to feature dependencies in practice
- 2 It requires very **few training examples** to estimate parameters
- 3 It is **computationally efficient** — ideal for real-time filtering

The Mathematical Formulation (Naïve Bayes)

For a given email $x = (x_1, \dots, x_d)$:

$$P(\text{spam} | x) \propto P(\text{spam}) \prod_{j=1}^d P(x_j | \text{spam})$$

Parameters estimated from training data:

$$P(x_j | \text{spam}) = \frac{\text{count}(x_j \text{ in spam}) + \alpha}{\text{total spam words} + \alpha \cdot V}$$

(α is Laplace smoothing, V is vocabulary size) **Key**

Innovation: The filter **adapts** to new spam tactics automatically as new labeled emails arrive.

Spam Filtering: Lessons for ML Practitioners

Why Spam Filters Are a Benchmark ML Success

- **Clear problem formulation:** Binary classification with easily measured performance
- **Abundant labeled data:** Users naturally label emails (by reporting spam)
- **Clear business value:** Saves billions of dollars in lost productivity annually
- **Continuous improvement:** Spammers adapt → models must adapt
- **Asymmetric error costs:** a **false positive** (legitimate mail sent to the spam folder) hurts far more than a **false negative** (spam in the inbox), so filters are tuned for **high precision** — formalised in Week 4

Key Takeaway for ML Design

The success of spam filtering demonstrates that **simple algorithms + high-quality features + abundant data** often outperform complex algorithms with limited data.

Modern Evolution

Spam filters now use **large language models (LLMs)** to understand context (e.g., distinguishing legitimate marketing emails from phishing scams). However, the core ML pipeline—labeled data → features → training → deployment → feedback loop—remains exactly as Mitchell described it.

Guided Discussion & Live Exercise

Class Discussion Questions

Instructions

Discuss in pairs (5 minutes), then share with class. There are no single correct answers—justify your position with evidence from today's papers.

- Q1. Samuel's Checkers program used **self-play** to generate experience. How does this differ from supervised learning with human-labeled data? What are the advantages and disadvantages of each approach?
- Q2. Jordan & Mitchell (2015) frame machine learning as a **new engineering discipline**. In what ways is ML similar to traditional engineering (civil, mechanical)? In what ways is it fundamentally different?
- Q3. Spam filters transitioned from rule-based systems to ML systems. Can you identify another domain where a similar transition is currently happening? What are the barriers to adoption?

Instructions

Individual work, 5 minutes. Submit on paper or LMS.

Exercise A: Formulate T-P-E

For each of the problems below, identify:

- ① **Task (T)** — what is being predicted?
- ② **Experience (E)** — what data is available?
- ③ **Performance (P)** — how will success be measured?

Problems:

- a **Credit default prediction** (banking)
- b **Image captioning** (computer vision + NLP)
- c **Recommendation system** for a streaming service
- d **Protein structure prediction** (biotech)

*Marking note: full credit requires a **measurable** P — “works well” is not a performance measure.*

Exercise B (individual work, 5 minutes)

You are evaluating two models for a medical diagnosis task:

Model	Training accuracy	Test accuracy
Model A	95%	94%
Model B	99%	80%

Questions:

- 1 Which model shows the larger **generalization gap**? What does that indicate?
- 2 Which model is more likely to be **overfitting** (high variance)? Justify.
- 3 Which model would you deploy in a clinical setting? Why?
- 4 Name two concrete ways to improve the other model.

Hint

Compare each model's training score with its *test* score — the gap, not the raw accuracy, is the diagnostic. Then consider the consequences of a misdiagnosis in healthcare.

Research Articles

- Samuel, A. L. (1959). **Some Studies in Machine Learning Using the Game of Checkers.** *IBM Journal of Research and Development*, 3(3), 210–229.
- Jordan, M. I., & Mitchell, T. M. (2015). **Machine Learning: Trends, Perspectives, and Prospects.** *Science*, 349(6245), 255–260.
- Wolpert, D. H. (1996). **The Lack of A Priori Distinctions Between Learning Algorithms.** *Neural Computation*, 8(7), 1341–1390.
- Vapnik, V. N., & Chervonenkis, A. Y. (1971). **On the Uniform Convergence of Relative Frequencies of Events to Their Probabilities.** *Theory of Probability and Its Applications*, 16(2), 264–280.

Case Study & Further Reading

- Sahami, M. et al. (1998). **A Bayesian Approach to Filtering Junk E-Mail.** *AAAI Workshop on Learning for Text Categorization*.
- Mitchell, T. M. (1997). **Machine Learning.** McGraw-Hill. Ch. 1.
- Bishop, C. M. (2006). **Pattern Recognition and Machine Learning.** Springer. Ch. 1.
- Shalev-Shwartz, S., & Ben-David, S. (2014). **Understanding Machine Learning.** Cambridge. Ch. 1-2.

End of Week 1

CSE 815: Machine Learning

“The goal of machine learning is to turn data into insight.”

— Unknown

Next Week: Week 2 (Part 1 & Part 2)

Mathematical Foundations: Linear Algebra, Probability, and Optimization

Prerequisite review: Vectors, matrices, derivatives, Bayes' rule, gradient descent

Reading before Week 2: [Bishop] Appendix · [Murphy] Ch. 2